

Pragyan Thapaliya

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INTRODUCTION

I am a software developer who enjoys turning ideas into working products. My journey started with building simple websites and gradually evolved into exploring backend engineering, automation, AI, and real-time systems. I enjoy learning how large-scale applications work and applying those concepts by building projects, experimenting with new technologies, and solving practical problems.

SKILLS

Languages & Core: JavaScript, **TypeScript**, SQL, Solidity

Frontend Development: **React.js**, Tailwind CSS

Backend Development: Node.js, **Express.js**, **NestJS**, REST APIs, **GraphQL**

Databases & Storage: PostgreSQL, MongoDB, **Redis**

Real-time & Messaging: Socket.IO, WebRTC, **BullMQ**

Web3 & Blockchain: **Solidity**, EVM, **Smart Contracts**, Ethereum, Ethers.js

DevOps & Infrastructure: Docker, Git, GitHub Actions, Linux

AI & Automation: **LangChain**, **LangGraph**, n8n

EXPERIENCE

Freelancer

Remote

Full Stack developer

2024-Present

- Architected and delivered full-stack web applications for multiple clients, focusing on maintainable architecture, performance optimization, and user experience.
- Designed backend services, database structures, authentication workflows, and API integrations for real-world business requirements.
- Collaborated directly with clients to analyze requirements, plan features, and deliver **production-ready solutions**.

Scodus

OnSite

Full Stack Developer (Internship)

1.5 month

- Developed **internal automation tools** for Scodus platforms (Finora Restaurant and Finora Finance), including **AI-powered menu extraction**, finance data extraction, and **document processing workflows** to automate manual business operations.
- Built and integrated full-stack solutions involving backend APIs, database workflows, AI integrations, and data processing pipelines for real-world restaurant and financial use cases.
- Gained hands-on exposure to business operations through onsite troubleshooting, including **POS system setup**, printer configuration, and understanding the intersection of software, hardware, and daily business workflows.

PROJECTS

Native WebRTC Peer-to-Peer Video Calling Platform (React Js,Socketio,WebRTC,ExpressJS)

- Architected a real-time peer-to-peer video communication platform using native WebRTC APIs without third-party video providers.
- Implemented a signaling server using **Socket.IO** to manage peer discovery, **SDP exchange**, **ICE negotiation**, and connection lifecycle.

- Designed matchmaking and room management logic for connecting random users through WebRTC peer connections.
- Explored real-time communication architecture including **STUN/TURN servers**, NAT traversal, and scalable signaling design.

Real-Time DNS Sinkhole & Ad-Blocking Engine (Nodejs,TypeScript,UDP,Socket.io,React,Docker)

- Built a custom UDP DNS server that intercepts DNS queries, evaluates domains against an optimized ~80K-domain blocklist for O(1) lookups, and proxies allowed traffic to Cloudflare (1.1.1.1).
- Implemented an in-memory analytics engine and REST/WebSocket APIs to process live query logs, domain categorization, and continuous metrics..
- Developed a React monitoring dashboard with real-time Socket.IO events, interactive visualization charts, and searchable block logs.

LMS Website (MERN Stack)

- Built and deployed a production-grade Learning Management System (MERN Stack) used by **100+ active users**.
- Developed features including course creation, enrollment management, digital content delivery, and student progress tracking.
- Implemented JWT authentication, **role-based access control (RBAC)**, secure REST APIs, and **automated** email notifications.
- Integrated payment tracking workflows and designed a scalable backend following production-ready **deployment** practices.

AI-Powered Financial Reconciliation System (Nest Js)

- Developed an automated transaction reconciliation platform using rule-based matching **algorithms** and AI-assisted decision workflows.
- Designed a **multi-layer reconciliation engine** using transaction scoring based on amount, date, and description similarity.
- Integrated **Gemini AI** for intelligent reconciliation suggestions and structured output validation using Zod schemas.
- Developed backend services to automate financial data processing and **reduce manual verification effort**.

ACHIEVEMENTS

- **1st Runner-up** at Cyber Utsav Hackathon , recognized for problem-solving and rapid development skills
- Participated in 100x Hackathon, building real-world solutions under time constraints in a collaborative environment
- Contributed as a tutor and **content creator** at CodeGhar YouTube Channel, teaching **programming** fundamentals

EDUCATION

NAMI International School

Kathmandu,Nepal

Higher Secondary Education (Class 12), Science

2024 – 2026

- Graduated among the **Top 3 students in the batch** with a **3.92 GPA**
- Science stream with strong foundation in Mathematics, Computer Science, and Programming

Amrit Science Campus (ASCOL)

Kathmandu,Nepal

Bachelor of Science in Computer Science and Information Technology, Science

2026 – 2030

- First Year Student